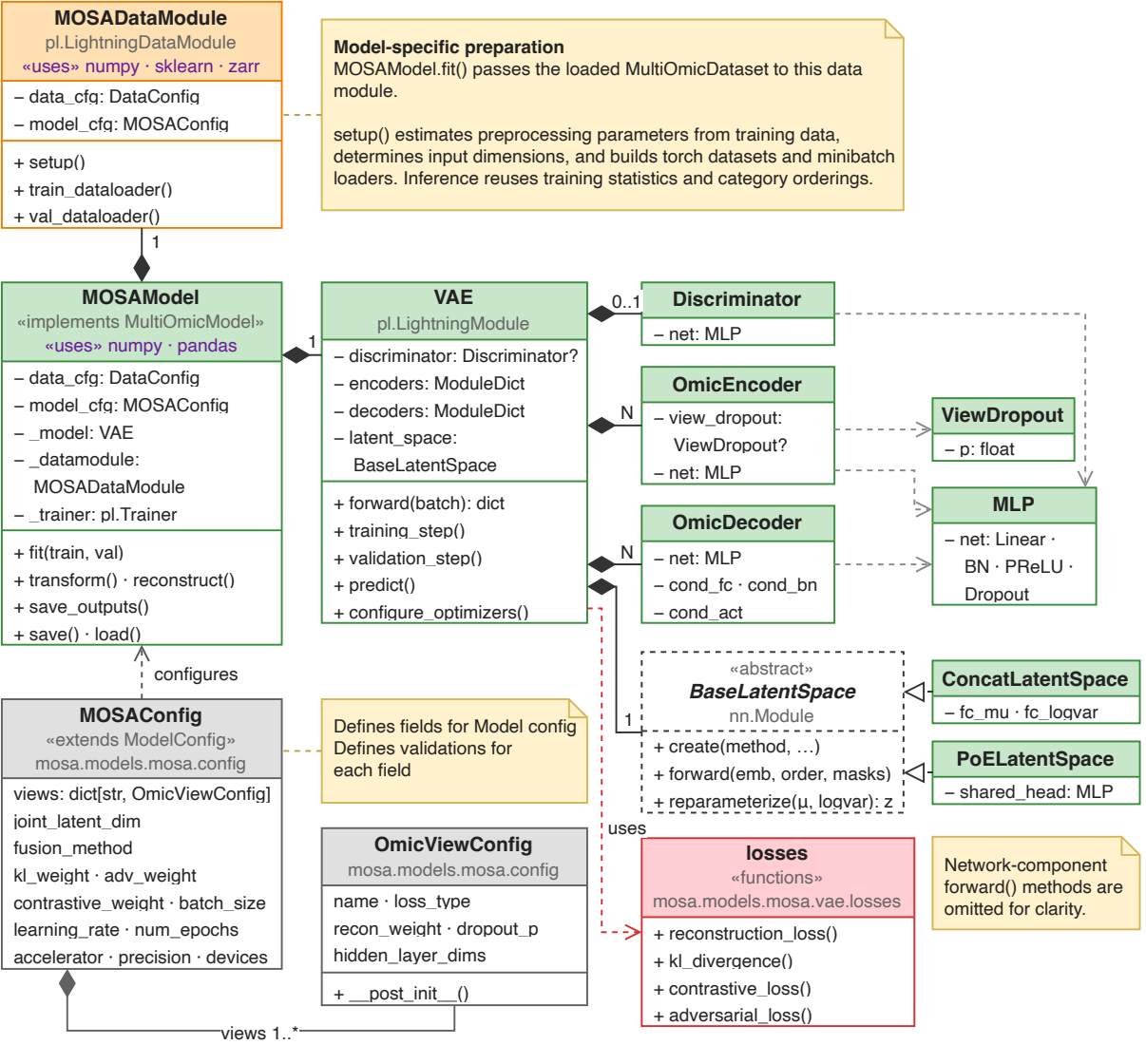
MOSADataModule
pl.LightningDataModule
«uses» numpy · sklearn · zarr
– data_cfg: DataConfig
– model_cfg: MOSAConfig
+ setup()
+ train_dataloader()
+ val_dataloader()
Model-specific preparation
MOSAModel.fit() passes the loaded MultiOmicDataset to this data module.
setup() estimates preprocessing parameters from training data, determines input dimensions, and builds torch datasets and minibatch loaders. Inference reuses training statistics and category orderings.
1
MOSAModel
«implements MultiOmicModel»
«uses» numpy · pandas
– data_cfg: DataConfig
– model_cfg: MOSAConfig
– _model: VAE
– _datamodule: MOSADataModule
– _trainer: pl.Trainer
+ fit(train, val)
+ transform() · reconstruct()
+ save_outputs()
+ save() · load()
1
VAE
pl.LightningModule
– discriminator: Discriminator?
– encoders: ModuleDict
– decoders: ModuleDict
– latent_space: BaseLatentSpace
+ forward(batch): dict
+ training_step()
+ validation_step()
+ predict()
+ configure_optimizers()
0..1
Discriminator
– net: MLP
N
OmicEncoder
– view_dropout: ViewDropout?
– net: MLP
ViewDropout
– p: float
MLP
– net: Linear · BN · PReLU · Dropout
N
OmicDecoder
– net: MLP
– cond_fc · cond_bn
– cond_act
configures
MOSAConfig
«extends ModelConfig»
mosa.models.mosa.config
views: dict[str, OmicViewConfig]
joint_latent_dim
fusion_method
kl_weight · adv_weight
contrastive_weight · batch_size
learning_rate · num_epochs
accelerator · precision · devices
Defines fields for Model config
Defines validations for each field
«abstract»
BaseLatentSpace
nn.Module
+ create(method, …)
+ forward(emb, order, masks)
+ reparameterize(μ, logvar): z
1
ConcatLatentSpace
– fc_mu · fc_logvar
PoELatentSpace
– shared_head: MLP
uses
OmicViewConfig
mosa.models.mosa.config
name · loss_type
recon_weight · dropout_p
hidden_layer_dims
+ __post_init__()
losses
«functions»
mosa.models.mosa.vae.losses
+ reconstruction_loss()
+ kl_divergence()
+ contrastive_loss()
+ adversarial_loss()
Network-component forward() methods are omitted for clarity.
views 1..*